

Wireshark: Essential for a Network Professional's Toolbox

This article, the second in the series, presents further experiments with Wireshark, the open source packet analyser. In this part, Wireshark will be used to analyse packets captured from an Ethernet hub.



The first article in the Wireshark series, published in the July 2014 issue of *OSFY*, covered Wireshark architecture, its installation on Windows and Ubuntu, as well as various ways to capture traffic in a switched environment. Interpretation of DNS and ICMP Ping protocol captures was also covered. Let us now carry the baton forward and understand additional Wireshark features and protocol interpretation.

To start with, capture some traffic from a network connected to an Ethernet hub—which is the simplest way to capture complete network traffic.

Interested readers may purchase an Ethernet hub from a second hand computer dealer at a throwaway price and go ahead to capture a few packets in their test environment. The aim of this is to acquire better hands-on practice of using

Wireshark. So start the capture and once you have sufficient packets, stop and view the packets before you continue reading.

An interesting observation about this capture is that, unlike only broadcast and host traffic in a switched environment, it contains packets from all source IP addresses connected in the network. Did you notice this?

The traffic thus contains:

- Broadcast packets
- Packets from all systems towards the Internet
- PC-to-PC communication packets
- Multicast packets

Now, at this point, imagine analysing traffic captured from hundreds of computers in a busy network—the sheer volume of captured packets will be baffling. Here, an important Wireshark